



Introduction to Data Science

Assignment 0

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Introduction

In this assignment, we will explore and implement key statistical concepts that are essential for analyzing data, making inferences, and drawing conclusions. These concepts play a critical role in research and real-world applications.

Tasks

Roulette Simulation and Profit Analysis

Roulette is a popular casino game played with a wheel that has numbered slots colored red, black, or green. In American roulette, the wheel has 38 slots: 18 red slots, 18 black slots, and 2 green slots labeled "0" and "00".

Players can place various types of bets, including betting on whether the outcome will be a red or black slot. In this exercise, we focus on a simple bet: betting on black.



If you place a bet on black and the outcome is indeed black, you win and double your money. However, if the outcome is red or green, you lose the amount you bet. For example, if you bet 1 dollar on black and win, you gain 1 dollar. If you lose, you forfeit your 1-dollar bet.

Because there are three colors, the chance of landing on the black isn't exactly $\frac{1}{2}$, it is less, specifically

$$\frac{18}{38} = \frac{9}{19}.$$

Consider the following tasks to simulate this game and analyze the expected outcomes of betting on black:

1. Write a function that simulates this game for N rounds, where each round consists of betting 1 dollar on black. The function should return your total earnings S_N after N rounds.
2. Use Monte Carlo simulation to study the distribution of total earnings S_N for $N = 10, 25, 100, 1000$. For each N , simulate 100,000 rounds and plot the distribution of total earnings. Analyze whether the distributions appear similar to a normal distribution and observe how the expected values and standard deviation change with N .
3. Repeat the previous simulation but for the average winnings $\frac{S_N}{N}$ instead of S_N . For each N , plot the distribution of average winnings and examine the changes in expected values and standard deviations with different values of N ($N = 10, 25, 100, 1000$).
4. Calculate the theoretical expected values and standard deviations of S_N for each N , and compare these theoretical values with your Monte Carlo simulation results. Report any differences between the theoretical and simulated values for each N .
5. Use the Central Limit Theorem (CLT) to approximate the probability that the casino loses money when you play $N = 25$ rounds, and verify this approximation using a Monte Carlo simulation.
6. Plot the probability that the casino loses money as a function of N for values N ranging from 25 to 1000. Discuss why casinos might encourage players to continue betting in light of these results.

Questions

1. Calculate the expected value of a single \$1 bet on black. Why is it impossible for a player to be profitable in the long run, and how does the casino guarantee its profit?
2. In part 3 of the task, you plotted the distribution of average winnings $\frac{S_N}{N}$. As N increases from 10 to 1000, what happens to the spread (variance) of this distribution, and which fundamental theorem of probability explains this behavior?
3. A data scientist collects a massive sample of $n = 10,000$ users to measure their daily time spent on an app. A junior colleague looks at the data and argues: "Because our sample size is so large, the standard deviation of our data must be very close to zero.". Explain the fundamental statistical flaw in this reasoning by distinguishing between standard deviation and standard error. What actually happens to these two metrics as $n \rightarrow \infty$?
4. You are analyzing Customer Lifetime Value (CLTV) for a mobile game. Revenue data is notoriously heavy-tailed and extremely right-skewed (i.e., 99% of players spend nothing, but 1% spend thousands of dollars). You take a random sample of $n = 35$ players, calculate the mean, and construct a 95% confidence interval using the Normal model, citing the widely used " $n > 30$ " rule for the Central Limit Theorem. Why is this statistical inference likely flawed?

Predicting the 2016 US Presidential Election Using Poll Aggregation

In 2012, statisticians and data scientists like Nate Silver demonstrated that aggregating multiple polls could produce remarkably accurate election forecasts. By combining results from different polling organizations, they reduced individual poll biases and sampling errors, creating more reliable predictions than any single poll could provide.

In this exercise, we apply similar statistical methods to analyze polling data from the 2016 US presidential election between Donald Trump and Hillary Clinton.



The data for this exercise is in a CSV file named `2016-general-election-trump-vs-clinton.csv`. Note that some rows represent subgroups (e.g., voters affiliated with specific parties) and contain NaN values in the Number of Observations column. These rows should be excluded to avoid errors.

Alright, let's roll up our sleeves and dive into the tasks!

Questions

0. The Central Limit Theorem (CLT) is fundamental to poll aggregation. Explain:
 - (a) Why does the CLT allow us to use normal approximations for sample proportions?
 - (b) What conditions must be satisfied for the CLT to apply to polling data?
 - (c) How does aggregating multiple polls improve upon the CLT's guarantees compared to using a single poll?

1. In election polling, we want to estimate the true proportion (p) of voters who support a candidate. We can't ask everyone, so we take a sample of N voters.

Let X_i represent the i -th voter's preference:

- $X_i = 1$, if they support the Democratic candidate
- $X_i = 0$, if they support the Republican candidate

The sample proportion $\hat{p} = \frac{1}{N} \sum_{i=1}^N X_i$ is our best guess for p , the proportion of voters who support the Democratic candidate. The Central Limit Theorem (CLT) states for large N , the distribution of $\hat{p}$ is approximately normal:

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{N}\right)$$

Based on the CLT result, derive the 95% confidence interval (CI) for p .

- a) The standard error formula $\sqrt{\frac{p(1-p)}{N}}$ contains p , which we don't know. Why do we use $\hat{p}$ instead? What's the impact on the CI?
 - b) The 95% CI uses $z = 1.96$. Where does this number come from? What would change for a 90% or 99% CI?
 - c) If N doubles, how does the CI width change? If p is close to 0.5 vs. close to 0 or 1, how does this affect the CI width?
2. Suppose the true population proportion is $p = 0.47$ (47% support for the Democrat). Perform a Monte Carlo simulation with $N = 30$ and 10^5 iterations to show that the CI derived in Question 1 captures true proportion p approximately 95% of the time.
 - a) In the simulation, you'll likely get coverage around 93-97%, not exactly 95%. Why? What factors affect how close it gets to 95%?
 - b) The simulation uses $N = 30$. Is this "large enough" for the CLT? What happens if you try $N = 10$ or $N = 100$?
 - c) The simulation fixes $p = 0.47$. Would coverage be different if $p = 0.10$ or $p = 0.90$? Why?
 - d) The frequentist interpretation: "If we repeated this process many times, 95% of CIs would contain p ." How is this different from "There's a 95% probability that p is in my specific CI"?
 3. Using the given dataset, make a DataFrame containing only the columns "Trump", "Clinton", "Pollster", "Start Date", "Number of Observations", and "Mode". Remember any rows where the Number of Observations is missing, must be excluded .

- a) If we decided NOT to exclude rows with missing Number of Observations, what alternative methods could we use? For each method, describe how you would implement it and discuss its advantages and disadvantages.
 - b) Why is Number of Observations crucial for calculating standard errors?
4. Create a time-series plot of poll results showing support percentages for Trump and Clinton, using different colors for each candidate. Include a smooth trend line to visualize support trends over time.
 - a) Why do we smooth the data with a moving average? Explain how the choice of window size (30 days) represents a specific point on the bias-variance tradeoff spectrum. What visual artifacts would appear with too-small or too-large windows?
 - b) If two polls on the same day show different results, how should we handle this in our visualization?
 - c) Should we weight polls by sample size when calculating daily averages? Why or why not?
 - d) If Clinton's smoothed line crosses above Trump's, does this mean she's "statistically significantly" ahead? What additional analysis would you need?
5. Calculate the total number of voters observed by summing all poll observations in the dataset.
 - a) If the same person is polled twice (panel data), how does this affect our "total voters" count and subsequent statistical inferences?
 - b) We excluded polls with missing sample sizes. If these tend to be smaller, cheaper polls, what direction of bias might this introduce?
6. Calculate the overall estimated proportion of voters favoring Trump and Clinton. Display these estimates in a table.
 - a) Polls have different methodologies (IVR, online, live). How could we test for "mode effects"?
 - b) If Trump has 45% in early polls and 48% in late polls, does our pooled estimate represent his "current" support? What time-weighted approach might be better?
7. Using the aggregated data, compute the 95% confidence intervals for Trump and Clinton support proportions.
 - a) Why do we typically use $z = 1.96$ (normal distribution) for proportions instead of a t-distribution, and what are the required assumptions?
 - b) If polls are correlated over time (autocorrelated), how does this affect our CI coverage? Would our CI be too wide or too narrow, and how could we adjust for it?
8. For illustrative purposes, assume there are only two parties, and let p denote the proportion of voters supporting Clinton. Consequently, $1 - p$ represents the proportion supporting Trump. We define the "spread" as the difference in support between Clinton and Trump:

$$d = p - (1 - p) = 2p - 1$$

Using the aggregated poll data, we estimate p as $\hat{p}$. Therefore, the estimated spread d can be approximated as:

$$d \approx 2\hat{p} - 1$$

This also implies that the standard error of the spread is twice as large as the standard error for $\hat{p}$. So, our confidence interval for the spread d is:

$$CI \text{ for } d = (2\hat{p} - 1) \pm 1.96 \times (2 \times SE(\hat{p}))$$

where $SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{N}}$ is the standard error of $\hat{p}$.

- Calculate the 95% confidence interval for the spread d , using the formula provided above.
 - Conduct a hypothesis test to determine if the spread d is significantly different from zero.
- a) If we get $\hat{d} = 0.02$ with $p\text{-value} = 0.04$, what should we conclude?
- b) What if the “only two parties” assumption is violated in reality?

Drug Safety Test

The dataset comes from a randomized controlled drug trial conducted by a medical group and shared by Vanderbilt University Department of Biostatistics. The study evaluates drug safety by comparing a Drug and Placebo group while tracking adverse effects, vital signs, and lab measures. You will conduct hypothesis testing using t-tests to determine statistical differences between groups.

Columns Explanation:

- **age**: Age of the participant
- **sex**: Gender of the participant (**male** or **female**).
- **trx**: Treatment group:
 - **"Drug"** → Received the actual drug.
 - **"Placebo"** → Received a placebo (control group).
- **week**: Week number in the study.
- **wbc**: White blood cell count (WBC) measurement.
- **rbc**: Red blood cell count (RBC) measurement.
- **adverse_effects**: Presence of adverse effects (**Yes** or **No**).

- **num_effects**: Number of adverse effects experienced by the participant.
1. Load the drug_safety.csv into a Pandas DataFrame.
 2. Remove samples that contain nan whenever it is needed.
 3. Display basic statistics (e.g., mean, standard deviation) for numeric columns.
 4. Group the dataset by **trx** (Drug vs. Placebo) and summarize key statistics for **wbc**, **rbc**, and **num_effects**.
 5. Change **adverse_effects** column so that you can define **mean** for it.
 6. For each metric below, determine if they differ significantly between the Drug and Placebo groups?
 - a. mean white blood cell count
 - b. mean red blood cell count
 - c. mean num effects
 - d. mean adverse effect
- Formulate null and alternative hypotheses:
 - H_0 : There is no significant difference
 - H_1 : There is a significant difference ●
- perform an independent t-test.
- Interpret the p-value and state whether to reject or fail to reject the null hypothesis.
 - If we set the p-value significant level to 0.05, which tests will fail? What about 0.1? What does this significant level mean?
 - use **scipy** library for it and report:
 - What is the **alternative** argument and what did you choose for each metric, why?
 - What is the **equal_var** argument and what does it do?

Questions

- 1- What are type 1 and type 2 errors in the context of this drug safety study? Which one is more critical in this specific study?
- 2- The independent t-test relies on several assumptions, List the key assumptions required for the independent t-test, explain how violations of these assumptions (like non-normality or unequal variances) could affect the results.
- 3- This study uses a randomized controlled design. Why is random assignment important from a probability perspective? How does randomization help reduce confounding variables? If

randomization were not used, how would this affect the interpretation of probabilities and hypothesis test results?

- 4- The independent t-test comparing mean red blood cell counts between the Drug and Placebo groups produced a p-value = 0.0663, what does this result imply about the difference in RBC counts between groups?
- 5- An engineer is monitoring the pressure inside an oil pipeline. Due to varying flow rates and environmental conditions, the pressure in the pipeline fluctuates slightly with time. The true average pressure of the pipeline is unknown. Pressure measurements, $X_1, X_2, \dots, X_n$ satisfy the following model:

$$X_i = \mu + \epsilon_i$$

Where μ is the unknown true average pressure, and ϵ_i represents random error. The errors are i.i.d. with mean 0 and unknown standard deviation σ . The pipeline's pressure is measured 100 times. If we construct an approximate 95% confidence interval for μ , this interval was constructed for one of the following purposes. Indicate which is correct and explain why:

- To estimate the average of the 100 pressure measurements and give ourselves some room for error in the estimate.
- To estimate the true average pressure of the pipeline and give ourselves some room for error in the estimate.
- To provide a range in which 95 of the 100 pressure measurements are likely to have fallen.
- To provide a range in which 95% of all possible pressure measurements are likely to fall.

Notes

- Upload your work as a zip file in this format on the website: DS_CA0_[Std number].zip. If the project is done in a group, include all of the group members' student numbers in the name.
- If the project is done in a group, only one member must upload the work.